Brightwater

Technical assessment submission

The Meraki Group | 26 October 2026

Website, proposed telephony design, support-ticket response and development summary.

Live website: Open Brightwater

Source and README: GitHub repository

Staff workspace: Operations

Contents

Task 1 - live links and acceptance status
Task 2 - call flow, build sheet, client questions and draft email
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Delivery and verification

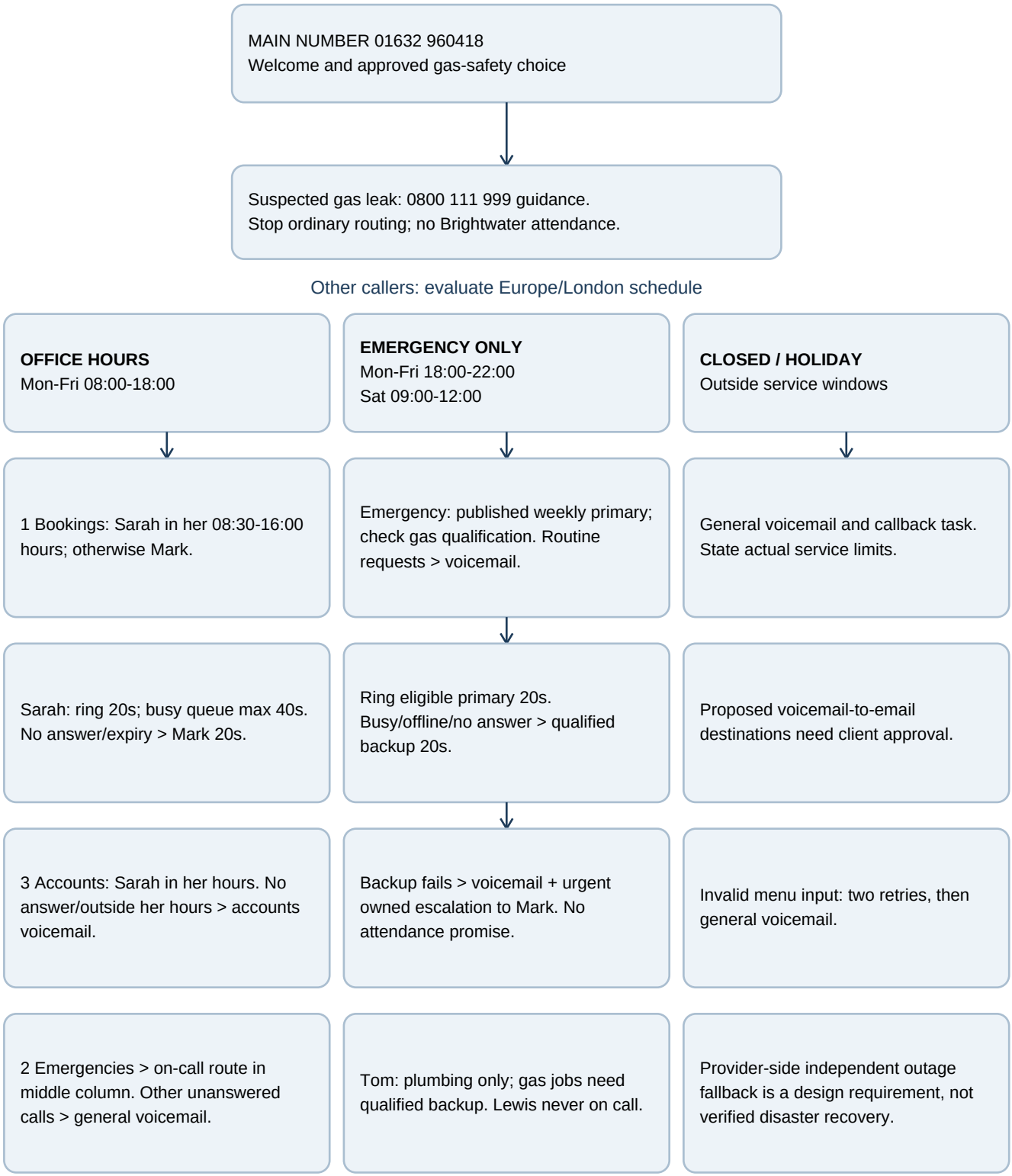
The owner confirmed successful website and call testing on a physical mobile device on 26 October 2026.

55 automated tests passed at the helpdesk release. The public site and workspace are deployed on Railway. The package brings together the assessment deliverables and a concise development summary.

The telephony build sheet presents the proposed client configuration. The additional Plivo pilot and Gmail helpdesk demonstrate the connected workflows. Client decisions are collected in the questions section.

Task 2 | Call-flow diagram

Proposed client routing; not a record of an implemented BT migration.



Task 2 | Build sheet and client communication

Person	Extension	Endpoint	Policy
Sarah Collins	201	Windows softphone + USB headset; optional supported desk phone	Bookings/accounts, weekdays 08:30-16:00
Mark Ellis	202	Supported iPad/mobile app	Bookings fallback; owner escalation; Gas Safe
Dan Hughes	203	Supported personal-mobile work app	On-call rota; Gas Safe
Priya Shah	204	Supported personal-mobile work app	On-call rota; Gas Safe
Tom Barker	205	Supported personal-mobile work app	On-call plumbing only; gas jobs to qualified backup
Lewis Grant	206	Optional supervised app, license subject to approval	Apprentice; never emergency on-call

Extensions are proposed, not existing facts. Confirm app/OS support, licenses, BYOD consent, battery permissions, mobile reception and headset needs. Do not assume the three existing BT cordless handsets support hosted SIP; inventory models first.

Object	Proposed setting
Main number	Port 01632 960418 only after account ownership, service association and port eligibility checks
Secondary number	01632 960419: do not cancel or port until its use, alarms/fax/dependencies and ownership are confirmed
Schedule	Europe/London with daylight-saving transitions; office and emergency windows as above
Holidays	Explicit approved dates; simulator closes the whole day; agree any live emergency exception separately
Welcome	Short brand welcome, then 1 bookings / 2 emergencies / 3 accounts
Invalid input	Two retries then general voicemail; verify provider timeout/retry semantics
Bookings group	Sarah, then Mark; Sarah omitted outside her working hours
Busy handling	Queue max 40 seconds; offer voicemail; expiry proceeds to fallback; confirm concurrency/licenses
Accounts	Sarah only during her hours; otherwise accounts voicemail
Emergency group	Current weekly primary, then qualified backup; no Lewis; Tom requires gas-qualified escalation
Ring timeout	20 seconds per destination, configurable 10-60 seconds in demo
Answer confirmation	Live provider should require engineer acceptance so personal carrier voicemail cannot consume escalation
General voicemail	Proposed info@brightwaterplumbing.co.uk, owned callback queue

Object	Proposed setting
Accounts voicemail	Proposed accounts@brightwaterplumbing.co.uk
Emergency failure	Urgent follow-up owned by Mark, with confirmed independent notification and acknowledgment policy
Outbound identity	Verified/authorized main number via provider app or supported callback; native mobile calls retain mobile identity
Call recording	Off by default; voicemail only until purpose, access, notice and retention are agreed
Outage	Provider-hosted announcement/fallback independent of website, plus alternate connectivity and documented escalation

Mailboxes are proposed destinations derived from the supplied address prefixes; confirm full domain, recipients, ownership and delivery controls. These client voicemail destinations remain proposals. Separately, the implemented helpdesk supports explicitly selected Gmail ticket replies; it does not activate these proposed voicemail destinations.

Questions before live configuration

1. Which Meraki platform, license package and mobile apps will be used? This determines queue, failover, acceptance and caller-ID capabilities.

2. Who covers the 08:00-08:30 and 16:00-18:00 gaps, accounts absences, lunches and leave? Demo assumes Mark for bookings and voicemail for accounts.

3. Who is each week's backup, who publishes the rota, and how do both acknowledge handover? Confirm qualified cover when Tom is primary and what happens if the backup cannot attend.

4. Who receives urgent alerts, by which independent channel, and who acts when no acknowledgment arrives? Agree a realistic threshold before configuring it.

5. Are the 20-second rings and 40-second queue acceptable? Confirm simultaneous calls, waiting prompts and voicemail escape key.

6. What happens on bank holidays and planned closures? Demo's full-day closure is only a proposal.

7. Confirm both BT numbers, account holder, billing address, contract/port details and the second number's purpose. Broadband is on the main line and ends March 2027; porting may affect connectivity or fees.

8. Confirm mailbox addresses, access, retention, consent/notice and whether recordings or transcriptions are wanted. Never email sensitive recordings to an unconfirmed group.

9. Confirm handset models, engineer mobile OS/devices, reception, app permissions, Wi-Fi and mobile data; agree BYOD/work-hours arrangements.

10. Confirm the port window, temporary number, fallback contacts, provider rollback options and acceptance-test sign-off. Do not cancel BT before a coordinated migration.

Draft email to Sarah - not sent

Hi Sarah,

The proposed system gives callers three choices: bookings, emergencies and accounts. Bookings reach you during your working hours, then Mark if you cannot answer. A short queue will help when you're already on a call. Accounts reach you or their own voicemail.

Emergency calls go to that week's engineer, then an agreed backup. If neither answers, Mark gets an urgent follow-up. Tom's gas-related jobs need a Gas Safe engineer. Outside the agreed service hours callers can leave a message, with clear guidance that emergency attendance is unavailable.

Engineers will use the phone-system app to call with the office number. Calling from the ordinary mobile dialer will still show their personal number.

Before setup, please confirm who covers your early/late hours and holidays; who approves and backs up the weekly rota; and who receives voicemail and urgent alerts. We'll also need the BT account details and the purpose of the old fax number. Because broadband shares the main line, we'll agree the migration with BT and our provider before moving anything.

We'll test the new system on a test number first and agree the transfer date with you.

Thanks

Task 3 | Investigation of ticket #10482

Treat Saturday emergency failure as high priority and Mark's caller ID as a linked, separate work item. Neither is marked fixed without evidence. The simulator reproduces possible routes; it does not establish the actual cause.

1. Obtain the three call times, caller number securely, number dialed and timezone; confirm whether the customer still needs help through the client's established process. Do not promise an engineer or a same-day technical repair.

2. Match provider inbound logs and call IDs. Verify calls reached the expected number/tenant, not an old answerphone or redirect.

3. Inspect Saturday 09:00-12:00 schedule, timezone, holidays and the effective configuration version. A wrong schedule could send callers to voicemail before the rota is read.

4. Compare published rota, acknowledgment and audit history to Priya's intended assignment. Check whether routing used a stale group, wrong extension, missing backup or unacknowledged handover.

5. Inspect each ring leg: destination, offer time, response, timeout and final disposition. Distinguish an unoffered call from a device that never notified, and provider voicemail from personal carrier voicemail.

6. With Priya, check registration, app sign-in, push/notification permissions, Focus/DND, battery restrictions, network/data reception and app version. Check business hours/presence on the provider as well as the handset.

7. Inspect backup route, queue overflow, voicemail destination and alerts. Confirm Mark actually received and acknowledged the fallback task; an emailed voicemail alone is insufficient for an urgent route.

8. Reproduce on a test number with Priya and the backup: answered, declined, busy, offline, no-answer and carrier-voicemail cases. Record timestamps and call IDs. Change one cause at a time and retain the previous configuration.

9. For Mark separately, establish whether he used the provider app or native dialer. Check verified outbound CLI, permissions, number assignment and provider/carrier restrictions, then test to an external phone. Never suggest arbitrary spoofing.

Information needed: exact times and numbers; Priya's expected rota, device/OS/app, network and presence; Mark's calling method and example destination/time; relevant call IDs, effective configuration, recent changes and alert delivery evidence. Collect personal data through a secure support channel.

Task 3 | Initial response and prevention

Initial reply, within the hour - draft, not sent:

Hi Sarah, I'm sorry those emergency calls did not reach Priya. I'm treating this as high priority and checking the Saturday call records, rota and Priya's app together. Please send the call times and the number the customer dialed through our secure support channel, and let us know whether the customer still needs assistance. I'll agree a temporary cover route with you while we investigate and update you within an hour, even if the cause is still being checked. I have also logged Mark's caller-ID issue separately so it is not lost behind the emergency fault. We will test both fixes with you before closing them.

Prevention after diagnosis: acknowledged primary/backup handover; Friday test call; qualified backup coverage; alerts for unregistered destinations and unanswered emergencies; owned follow-ups with acknowledgment; versioned configuration and rollback; scheduled test calls for Saturday/holiday/DST boundaries; app onboarding and BYOD guidance. Add tests for the actual root cause once known, rather than declaring every possible failure fixed.

Task 4 | Written summary (1 of 2)

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Approach and priorities

Development began with the customer journey: clear service information, published prices, coverage rules, emergency guidance and a straightforward enquiry form. Next.js and TypeScript provided a consistent foundation for the website and its supporting application. The assessment deliverables were then extended with a staff workspace, task management, automated follow-up and a voice-call demonstration.

The public website contains Home, Services, Pricing, Coverage, About, Contact and Landlords pages. Enquiries receive a reference and are stored for staff follow-up. The workspace brings together individual accounts, assignments, priorities, rota management, audit history and operational reporting. The support desk adds customer conversations, private portal links, Gmail ticket replies and response deadlines. Plivo provides the inbound telephone connection, with ElevenLabs speech and DeepSeek responses.

Decisions and assumptions

The website and workspace share a Railway web service. PostgreSQL, Redis, the worker and scheduler run as separate services. PostgreSQL holds business records; Redis schedules background work. Transactional checks and stable job identifiers prevent duplicate processing. SQLite supports local development.

The voice agent retrieves business information through defined tools and records its actions in the conversation map. It identifies itself as AI. Booking and dispatch proposals remain subject to operator review within the simulation; telephone audio and explicitly selected support emails use connected providers. This separation keeps demonstration workflows distinct from customer-facing delivery.

The assessment's team, prices, postcode restrictions and gas-emergency guidance informed the implementation. Illustrative imagery and a temporary wordmark support the current presentation. The telephony build sheet identifies proposed settings and client decisions, including rota ownership, qualified backup, voicemail destinations and the effect of number porting on the existing broadband service.

Quality checks

At the helpdesk release, 55 automated tests passed. Coverage includes routing, postcode eligibility, permissions, session revocation, task ownership, stale-update handling, automation idempotency, ticket privacy and portal access. Production builds and deployed service health were checked. Database migration and an isolated snapshot restore were verified.

Browser checks covered desktop and mobile layouts, enquiry and workspace interactions, and the ticket workflow. Successful website and call testing on a physical mobile device was confirmed by the project owner on 26 October 2026. A separate provider-stream test exercised greeting audio, interruption handling and a subsequent response. The conversation map also passed a synthetic gesture stress test.

Task 4 | Written summary (2 of 2)

Tools and AI assistance

The implementation uses Next.js, React, TypeScript, Node.js, PostgreSQL, SQLite, Redis/BullMQ, Docker and Railway. Plivo, DeepSeek and ElevenLabs provide telephony and conversational services; Nodemailer, ImapFlow and Mailparser support the Gmail integration.

ChatGPT/Codex supported planning, implementation, debugging, tests and documentation. Taste Skill and Impeccable informed interface work, and gstack supported browser verification. AI image generation supplied design concepts and labelled illustrative photography. Generated material required factual review, and browser testing identified integration issues beyond the automated test coverage.

Learning and improvements

Several issues shaped the final implementation: staff selectors were moved to a shared directory so new engineers appear consistently; microphone permissions were corrected; origin checks were aligned with deployment settings; and map updates were bounded and scheduled by animation frame. Voice-turn handling was adjusted to accommodate pauses and avoid ending calls after unrelated questions or a simple thank-you.

These changes reinforced the value of testing the complete user journey alongside individual functions. Clear distinctions between a queued action, provider acceptance and a customer outcome also improved the delivery tracking and documentation.

Remaining assessment items

The next handover steps are to obtain the original Brand Pack, resolve the free-hosting requirement through a qualifying website deployment or an agreed exception, and send the prepared PDF and repository link to Meraki. Broader production acceptance criteria are maintained separately in the operational readiness guide.